



Cong-Thanh Vu, Ph.D.

Director, Robotics and Autonomous Systems Laboratory (RASL)

Assistant Professor, Department of Mechatronics Engineering, HAUI, Vietnam

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Location: Hanoi, Vietnam

RESEARCH INTERESTS

Human-Robot Interaction & Social Robotics: Adaptive companionship, proactive assistance, human-aware mobile robot navigation, and group following.

Field & Legged Robotics: Agricultural mobile robots, adaptive pesticide spraying, passive dynamic walking, and humanoid locomotion.

Robot Learning & AI: Deep reinforcement learning, continual learning for tracking, and Vision-Language-Action (VLA) multimodal models.

EDUCATION

National Cheng Kung University (NCKU)

Ph.D. in Mechanical Engineering

Tainan, Taiwan

Mar. 2022 – Jun. 2026

Hanoi University of Industry (HAUI)

M.S. in Mechatronics Engineering

Hanoi, Vietnam

Aug. 2019 – Jun. 2021

Hanoi University of Industry (HAUI)

B.S. in Mechatronics Engineering

Hanoi, Vietnam

Aug. 2015 – Jun. 2019

ACADEMIC & PROFESSIONAL EXPERIENCE

Hanoi University of Industry (HAUI)

Assistant Professor & Lab Director (RASL), Department of Mechatronics Engineering

Hanoi, Vietnam

Aug. 2026 – Present

National Cheng Kung University (NCKU)

Graduate Research Assistant, Department of Mechanical Engineering

Tainan, Taiwan

Mar. 2022 – Jun. 2026

University of Economics - Technology for Industries (UNETI)

Research Assistant, Department of Mechanical Engineering

Hanoi, Vietnam

Aug. 2021 – Feb. 2022

Hanoi University of Industry (HAUI)

Graduate Research Assistant, Department of Mechanical Engineering

Hanoi, Vietnam

Aug. 2019 – Jun. 2021

HONORS & AWARDS

- **Full Scholarship**, National Cheng Kung University (NCKU), Taiwan (2022 – 2026)
- **Travel Award**, National Science and Technology Council (NSTC), Taiwan (2024)
- **Talent Scholarship**, Foxconn Technology Group, Vietnam (2017, 2018)

SELECTED PUBLICATIONS (14 PEER-REVIEWED PAPERS)

Refereed Journal Articles

1. **Towards Adaptive Interaction Strategies for Human Companion Robot via Deep Reinforcement Learning**
Cong-Thanh Vu, Yen-Chen Liu
IEEE Transactions on Systems, Man, and Cybernetics: Systems, accepted, 2026
2. **Context-Aware Adaptive Pesticide Spraying for Agricultural Robots under Changing Weather and Terrain Using Vision-Language Models**

Cong-Thanh Vu, Yen-Chen Liu

Computers and Electronics in Agriculture, Vol. 252, pp. 112092, Sep. 2026; ISSN 0168-1699; DOI 10.1016/j.compag.2026.112092

Refereed Conference Proceedings

1. **Adaptive Companionship for Group-Following Robots: Handling Dynamically Changing Group Formations**
Cong-Thanh Vu, Yen-Chen Liu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Pittsburgh, USA, accepted
2. **Gimbal-Based Human Tracking for Companion Robots Using Continual Learning**
Cong-Thanh Vu, Ching-Chieh Liu, Yen-Chen Liu
IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Genova, Italy, accepted
3. **Multimodal-Language-Model-Driven Interaction and Companionship for Service Robots in Elderly-Care Facilities**
Ching-Chieh Liu, **Cong-Thanh Vu**, Yen-Chen Liu
IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Genova, Italy, accepted
4. **Learning Diverse Humanoid Tasks via Synthetic Video Scenarios without Real World Data**
Yun-Hao Tsai, **Cong-Thanh Vu**, Yen-Chen Liu
IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Genova, Italy, accepted
5. **Autonomous Adjustment of Tracking Position in Dynamic Environments for Human-Following Robots Using Deep Reinforcement Learning**
Cong-Thanh Vu, Yen-Chen Liu
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025
6. **Learning Virtual Passive Dynamic Walking Using the Knead Walker Model for Guiding Policies**
Cong-Thanh Vu, Chi Cheng Lai, Yen-Chen Liu
IEEE-RAS International Conference on Humanoid Robots (Humanoids), 2025
7. **Autonomous Navigation for Human-Following Robots Based on Optimized Position Tracking**
Cong-Thanh Vu, Hsin-Hui Huang, Yen-Chen Liu
International Conference on Control and Robotics Engineering (ICCRE), 2025
8. **Toward Autonomous Navigation for Agriculture Robots in Orchard Farming**
Cong-Thanh Vu, Hsin-Chia Chen, Yen-Chen Liu
IEEE International Conference on Recent Advances in Systems Science and Engineering (RASSE), 2024
9. **Deep Reinforcement Learning for Multi-Robot Local Path Planning in Dynamic Environments**
Cong-Thanh Vu, Yen-Chen Liu
International Workshop on Robot Motion and Control (ROMOCO), 2024
10. **Mobile robot motion control using a combination of fuzzy logic method and kinematic model**
Anh-Tu Nguyen, **Cong-Thanh Vu**
Intelligent Systems and Networks (ICISN), 2022
11. **Obstacle avoidance for autonomous mobile robots based on mapping method**
Anh-Tu Nguyen, **Cong-Thanh Vu**
Advanced Mechanical Engineering, Automation and Sustainable Development (AMAS), 2022
12. **Development of a multiple-sensor navigation system for autonomous guided vehicle localization**

PROFESSIONAL SERVICES & ACTIVITIES

Journal Reviewer:

- *IEEE/ASME Transactions on Mechatronics* (TMECH)
- *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (TSMC)
- *Journal of the Franklin Institute* (JFI)

Conference Reviewer:

- *IEEE International Conference on Robotics and Automation* (ICRA)
- *IEEE/RSJ International Conference on Intelligent Robots and Systems* (IROS)
- *IEEE/ASME International Conference on Advanced Intelligent Mechatronics* (AIM)
- *IEEE International Conference on Systems, Man, and Cybernetics* (SMC)
- *IEEE-RAS International Conference on Humanoid Robots* (Humanoids)